Founding Team Member: Laser-Matter Interaction & Optics Scientist

Company: Cascade Dynamics (https://www.cascade-dynamics.com)

Status: Stealth Mode Deep-Tech Startup

Location: Lab-based deep tech hub in Champaign, IL, Boston, MA or Chicago, IL

Compensation: Meaningful Founding Equity Package (with transitioning base salary post funding round)

About the Company

We are a stealth-mode, pre-revenue deep-tech company engineering a new class of ion sources at the intersection of light, matter, plasma, and charged particle beams. Powered by a foundational patent portfolio, we are developing and testing a proprietary ion-source architecture intended to increase particle energy and reduce reliance on hazardous, high-maintenance, and supply-chain-vulnerable legacy technologies across multiple industrial and scientific applications.

Our founding team brings experience in plasma physics and entrepreneurship. We are seeking an experimentalist with expertise in laser–matter interactions to own the pulsed-laser, beam-delivery, and optical-diagnostic work required for Photon Released Ions from Suspended Matter (PRISM), our heavy-ion source. Our initial commercial focus is expanding access to heavy-ion single-event-effects testing. We have completed I-Corps customer discovery and have submitted an NSF Phase I SBIR proposal.

The Role

As a Founding Team Member, you will develop laser–matter experiments with founder and Chief Scientist Erik Ziehm and lead the laser and optical work. You will select and commission the appropriate pulsed-laser system, build the vacuum-compatible optical train, and work hands-on across integrated test campaigns from initial alignment through troubleshooting and repeatability testing.

You will work directly at the boundary between laser physics and ion-source development, integrating the laser and optical diagnostics into the experimental system. Working with Erik and the team, you will define optical access, alignment and timing requirements, and identify diagnostic needs. You will plan experiments and document settings and hardware changes.

Core Responsibilities

- **Pulsed-Laser Operations:** Select, commission, operate, and maintain the pulsed-laser system suited to the experiments. Establish reproducible operating conditions by measuring and stabilizing the laser output.
- **Calibrated Beam Delivery:** Design, align, and characterize the optical path through the chamber viewports to the suspended-particle interaction region. Measure delivered pulse energy, pulse duration, beam profile, focal-spot size and beam–particle overlap, fluence, repetition rate, pointing stability, timing and drift at the interaction region, and quantify measurement uncertainty.
- **Vacuum Optical Integration:** Specify and integrate windows, feedthroughs, focusing and steering arrangements, debris shields and replaceable components; establish cleaning procedures. Characterize transmission stability, deposition, contamination, thermal drift and optical damage limits; establish maintenance intervals.
- **Optical Diagnostics & Spectroscopy:** Implement calibrated imaging, scattering measurements, spectroscopy, photodiode measurements, or laser-induced fluorescence where justified to characterize the particle population, laser–particle overlap, material release and plasma response. Investigate laser absorption and interpret optical emission spectra. Correlate optical evidence with synchronized extracted-ion measurements.

- **Timing, Safety & Cross-Functional Execution:** Define laser trigger, delay, jitter, and acquisition requirements for integrated experiments. Establish compliant enclosures, shutters, beam dumps, access controls, safe states, and procedures; work with the team on synchronized acquisition and interlock implementation, and contribute supportable results to technical milestones, proposals, and patent disclosures.

Required Qualifications

- **Educational Background:** Ph.D., M.S., or equivalent experimental record in optics, photonics, applied physics, electrical engineering, or a closely related field.
- **Hands-On Laser Systems Experience:** Demonstrated experience commissioning, aligning, operating, maintaining, and troubleshooting pulsed-laser experiments and complex free-space optical systems. Experience studying laser absorption and interpreting optical emission spectra.
- **Beam Metrology & Experimental Rigor:** Experience calibrating optical diagnostics and measuring pulse energy, pulse duration, spatial profile, spot size, overlap, fluence, repetition rate, pointing, timing and drift at the interaction region, with quantified measurement uncertainty.
- **Vacuum & Harsh-Environment Optics:** Practical command of vacuum-compatible optical hardware, viewport fouling, contamination and debris control, optical damage, thermal effects, serviceability, and laboratory laser-safety practice.
- **Mindset:** A high-agency, startup-ready experimentalist committed to sustained hands-on laboratory work in a fast-paced, un-siloed "Skunkworks" environment. Performance dictates corporate progress here.
- **Security Clearance:** Preferred ability to obtain a U.S. Government Security Clearance (U.S. Citizenship typically required for downstream defense prime deployments).

Preferred Extras

- Experience with laser-produced plasmas, laser ion sources, suspended particles, aerosols, powders, droplets, or optical diagnostics in high-vacuum or magnetized-plasma systems.
- Experience with time-resolved imaging, scattering, laser-induced fluorescence, synchronized acquisition, contamination-prone optical systems, or interfaces to electron cyclotron resonance (ECR) ion sources and accelerators.

What We Offer

- **Direct Equity Ownership:** A true co-founding technical stake in a venture tackling a multi-billion-dollar bottleneck with definitive, validating customer pull.
- **High Impact Culture:** The autonomy to build laser and optical capabilities from scratch alongside world-class peers, completely free of corporate bureaucracy or traditional academic slow-downs.